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Roll No

MI-7005(1)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mine Economics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the differences between tenor, grade and specification of an Ore? Illustrate with example. 7
b) Is it possible to convert a mineral resource to a mineral reserve? If so, then how and under what conditions? 7
2. a) What do you understand by Area of Influence of a Borehole in assessing an ore body? 7
b) Describe a suitable method of sampling and respective details of how the weighted average grade is calculated for a final sample prepared from a drift face in underground? 7
3. a) What is Mine valuation? 2
b) Describe in details, various criteria-wise methods of Mine Valuation? 8
c) What is salting in Mine Valuation? How can it be prevented? 2+2=4

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PTO

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4. a) Under what conditions a Mining Company may need to wind up? 6
b) What are the precautions and notices/approvals needed for a Mine closure as per Mining legal provisions? 8
5. a) What do you mean by sales price per Unit of a product? What are the various cost elements included form its Prime cost level in it? Write the mathematical expressions. 6
b) Uniform Rectifier Company has the following details:-
Sales= Rs.1,20,000; Fixed Cost= Rs. 25,000; Variable Cost=Rs.45,000. 8
Find
i) Contribution
ii) Profit
iii) Break-even Point (BEP)
iv) Margin of Safety.
6. a) Differentiate between a Private Company and a Public Sector Unit. 6
b) Describe in details the procedure to be adopted for opening a new Company. 8
7. a) Define and express mathematically NPV in case of a capital Investment. 4
b) A state Govt. is planning a small Hydro project on a perennial stream. In addition to the production of electric power, this project will have different benefits as evaluated below. The other relevant details to decide about implementation of projects are : 10
Initial cost (CAPEX) = Rs. 8,00,00,000
Annual Power sales = Rs. 60,00,000

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Annual flood Control savings = Rs. 30,00,000

Annual Recreation benefits = Rs. 20,00,000

Annual irrigation benefits = Rs. 50,00,000

Annual operating & maintenance cost = Rs. 30,00,000

Life of project = 50 yrs

Value of Interest Rate, $i=12\%$

Decide based on Benefit-cost (BC) Ratio, whether project should be implemented or not.

8. Write short notes on any four of the following:- $4 \times 3.5 = 14$
- i) Difference between depreciation and amortization.
 - ii) Precautions in recovery and preservation of Cores of Boreholes.
 - iii) Sample Preparations methods.
 - iv) Difference between Depreciated Value and Salvage value of an Asset.
 - v) Difference between Prospecting and Exploration.
 - vi) Methods of Book Keeping.

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